Chapter 7

2.

a. The sample mean is the center of the interval, so $\bar{x} = \frac{114.4 + 115.6}{2} = 115$.

b. The interval (114.4, 115.6) has the 90% confidence level. The higher confidence level will produce a wider interval.

4.

a. $58.3 \pm \frac{1.96(3)}{\sqrt{25}} = 58.3 \pm 1.18 = (57.1, 59.5)$.

b. $58.3 \pm \frac{1.96(3)}{\sqrt{100}} = 58.3 \pm .59 = (57.7, 58.9)$.

c. $58.3 \pm \frac{2.58(3)}{\sqrt{100}} = 58.3 \pm .77 = (57.5, 59.1)$.

d. 82% confidence $\Rightarrow 1 - \alpha = .02 \Rightarrow \alpha = .18 \Rightarrow \alpha/2 = .09$, and $z_{.09} = 1.34$. The interval is $58.3 \pm \frac{1.34(3)}{\sqrt{100}} = (57.9, 58.7)$.

e. $n = \left[\frac{2(2.58)3}{1}\right]^2 = 239.62 \nearrow 240$.

10.

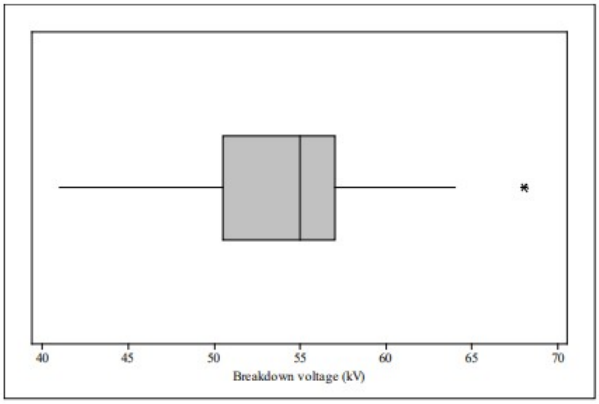
a. When $n = 15$, $2\lambda \sum X_i$ has a chi-squared distribution with 30 df. From the 30 df. row of Table A.6, the critical values that capture lower and upper tail areas of .025 (and thus a central area of .95) are 16.791 and 46.979. An argument parallel to that given in Example 7.5 gives $\left(\frac{2\sum x_i}{46.979}, \frac{2\sum x_i}{16.791}\right)$ as a 95% CI for $\mu = \frac{1}{\lambda}$. Since $\sum x_i = 63.2$ the interval is (2.69, 7.53).

b. A 99% confidence level requires using critical values that capture area .005 in each tail of the chi-squared curve with 30 df.; these are 13.787 and 53.672, which replace 16.791 and 46.979 in **a**.

c. $V(X) = \frac{1}{\lambda^2}$ when X has an exponential distribution, so the standard deviation is $\frac{1}{\lambda}$, the same as the mean. Thus the interval of **a** is also a 95% CI for the standard deviation of the lifetime distribution.

16.

a. The boxplot shows a high concentration in the middle half of the data (narrow box width). There is a single outlier at the upper end, but this value is actually a bit closer to the median (55 kV) than is the smallest sample observation.



b. From software, $\bar{x} = 54.7$ and $s = 5.23$. The 95% confidence interval is then

$$\bar{x} \pm 1.96\frac{s}{\sqrt{n}} = 54.7 \pm 1.96\frac{5.23}{\sqrt{48}} = 54.7 \pm 1.5 = (53.2,\ 56.2)$$

We are 95% confident that the true mean breakdown voltage under these conditions is between 53.2 kV and 56.2 kV. The interval is reasonably narrow, indicating that we have estimated μ fairly precisely.

c. A conservative estimate of standard deviation would be (70 – 40)/4 = 7.5. To achieve a margin of error of at most 1 kV with 95% confidence, we desire

$$1.96\frac{s}{\sqrt{n}} \le 1 \Rightarrow n \ge \left[\frac{1.96s}{1}\right]^2 = \left[\frac{1.96(7.5)}{1}\right]^2 = 216.09.$$

Therefore, a sample of size at least $n = 217$ would be required.

20.

a. With the numbers provided, a 99% confidence interval for the parameter p is

$$\frac{.53 + \frac{(2.58)^2}{2(2343)} \pm 2.58\sqrt{\frac{(.53)(.47)}{2343} + \frac{(2.58)^2}{4(2343)^2}}}{1 + \frac{(2.58)^2}{2343}} = (.503,\ .556).$$

We are 99% confident that the proportion of <u>all</u> adult Americans who watched streamed programming was between .503 and .556.

b. $n = \left[\frac{2z_{\alpha/2}\hat{p}\hat{q}}{w}\right]^2 = \left[\frac{2(2.58)(.5)(.5)}{.05}\right]^2 = 665.64$, so n must be at least 666.

26. Using the quadratic equation to solve the limits $\frac{\overline{x}-\mu}{\sqrt{\mu/n}} = \pm z$ gives the solutions

$\mu = \left(\overline{x} + \frac{z^2}{2n}\right) \pm z \cdot \frac{\sqrt{z^2 + 4n\overline{x}}}{2n}$. For the data provided, n = 50 and $\overline{x}$ =203/50 = 4.06. Substituting these and using z = 1.96, we are 95% confident that the true value of μ is in the interval

$\left(4.06 + \frac{1.96^2}{2(50)}\right) \pm 1.96 \cdot \frac{\sqrt{1.96^2 + 4(50)(4.06)}}{2(50)}$ = 4.10 ± .56 = (3.54, 4.66).

Note that for large n, this is approximately equivalent to $\overline{x} \pm z\frac{\sqrt{\overline{x}}}{\sqrt{n}} = \hat{\mu} \pm z\frac{\hat{\sigma}}{\sqrt{n}}$ (since $\mu = \sigma^2$ for the Poisson distribution).

30.

a. $t_{.025,10} = 2.228$

b. $t_{.025,15} = 2.131$

c. $t_{.005,15} = 2.947$

d. $t_{.005,4} = 4.604$

e. $t_{.01,24} = 2.492$

f. $t_{.005,37} \approx 2.712$

34. $n = 14,\ \overline{x} = 8.48$, $s = .79$; $t_{.05,13} = 1.771$

a. A 95% lower confidence bound: $8.48 - 1.771\left(\frac{.79}{\sqrt{14}}\right) = 8.48 - .37 = 8.11$. With 95% confidence, the value of the true mean proportional limit stress of all such joints is greater than 8.11 MPa. We must assume that the sample observations were taken from a normally distributed population.

b. A 95% lower prediction bound: $8.48 - 1.771(.79)\sqrt{1 + \frac{1}{14}} = 8.48 - 1.45 = 7.03$. If this bound is calculated for sample after sample, in the long run 95% of these bounds will provide a lower bound for the corresponding future values of the proportional limit stress of a single joint of this type.

36. $n = 26,\ \overline{x} = 370.69$, $s = 24.36$; $t_{.05,25} = 1.708$

a. A 95% upper confidence bound: $370.69 + (1.708)\left(\frac{24.36}{\sqrt{26}}\right) = 370.69 + 8.16 = 378.85$

b. A 95% upper prediction bound: $370.69 + 1.708(24.36)\sqrt{1 + \frac{1}{26}} = 370.69 + 42.45 = 413.14$

c. Following a similar argument as the one presented in the Prediction Interval section, we need to find the variance of $\overline{X} - \overline{X}_{\text{new}}$: $V\left(\overline{X} - \overline{X}_{\text{new}}\right) = V\left(\overline{X}\right) + V\left(\overline{X}_{\text{new}}\right) = V\left(\overline{X}\right) + V\left(\tfrac{1}{2}(X_{27} + X_{28})\right)$

$= V\left(\overline{X}\right) + V\left(\tfrac{1}{2}X_{27}\right) + V\left(\tfrac{1}{2}X_{28}\right) = V\left(\overline{X}\right) + \tfrac{1}{4}V\left(X_{27}\right) + \tfrac{1}{4}V\left(X_{28}\right)$

$= \frac{\sigma^2}{n} + \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \sigma^2\left(\frac{1}{2} + \frac{1}{n}\right)$. We eventually arrive at $T = \frac{\overline{X} - \overline{X}_{\text{new}}}{s\sqrt{\frac{1}{2} + \frac{1}{n}}} \sim t$ distribution with $n - 1$ df, so the new prediction interval is $\overline{x} \pm t_{\alpha/2,n-1} \cdot s\sqrt{\tfrac{1}{2} + \tfrac{1}{n}}$. For this situation, we have

$370.69 \pm 1.708(24.36)\sqrt{\frac{1}{2} + \frac{1}{26}} = 370.69 \pm 30.53 = (340.16, 401.22)$.

42.

a. $\chi^2_{.1,15} = 22.307$ (.1 column, 15 df row)

b. $\chi^2_{.1,25} = 34.381$

c. $\chi^2_{.01,25} = 44.313$

d. $\chi^2_{.005,25} = 46.925$

e. $\chi^2_{.99,25} = 11.523$ (.99 col., 25 df row)

f. $\chi^2_{.995,25} = 10.519$

44. $n - 1 = 8$, $\chi^2_{.025,8} = 17.534$, $\chi^2_{.975,8} = 2.180$, so the 95% interval for σ^2 is $\left(\frac{8(7.90)}{17.534}, \frac{8(7.90)}{2.180}\right) = (3.60, 28.98)$.

The 95% interval for σ is $\left(\sqrt{3.60}, \sqrt{28.98}\right) = (1.90, 5.38)$.

46.

a. Using a normal probability plot, we ascertain that it is plausible that this sample was taken from a normal population distribution.

b. With $s = 1.579$, $n = 15$, and $\chi^2_{.95,14} = 6.571$, the 95% upper confidence bound for σ is

$$\sqrt{\frac{14(1.579)^2}{6.571}} = 2.305.$$

Chapter 8

4. We reject H_0 iff P-value $\leq \alpha$.

a. Do not reject H_0, since $.084 > .05$.

b. Do not reject H_0, since $.003 > .001$.

c. Do not reject H_0, since $.498 > .05$.

d. Reject H_0, since $.084 \leq .10$.

e. Do not reject H_0, since $.039 > .01$.

f. Do not reject H_0, since $.218 > .10$.

8. H_0: $\mu = 40$ v. H_a: $\mu \neq 40$, where μ is the true average burn-out amperage for this type of fuse. The alternative reflects the fact that a departure from $\mu = 40$ in either direction is of concern. A type I error would say that one of the two concerns exists (either $\mu < 40$ or $\mu > 40$) when, in fact, the fuses are perfectly compliant. A type II error would be to fail to detect either of these concerns when one exists.

10. Let μ_1 = the average amount of warpage for the regular laminate, and μ_2 = the analogous value for the special laminate. Then the hypotheses are H_0: $\mu_1 = \mu_2$ v. H_a: $\mu_1 > \mu_2$. Type I error: Conclude that the special laminate produces less warpage than the regular, when it really does not. Type II error: Conclude that there is no difference in the two laminates when in reality, the special one produces less warpage.

12.

a. H_0: $\mu = 1300$ v. H_a: $\mu > 1300$.

b. When H_0 is true, $\overline{X}$ is normally distributed with mean $E(\overline{X}) = \mu = 1300$ and standard deviation $\frac{\sigma}{\sqrt{n}} = \frac{60}{\sqrt{10}} = 18.97$. Values more contradictory to H_0 (more indicative of H_a) than $\overline{x} = 1340$ would be anything above 1340. Thus, the P-value is

P-value = $P(\overline{X} \geq 1340$ when H_0 is true) = $P\left(Z \geq \frac{1340-1300}{18.97}\right) = 1 - \Phi(2.11) = .0174$.

In particular, since $.0174 > .01$, H_0 would <u>not</u> be rejected at the $\alpha = .01$ significance level.

c. When $\mu = 1350$, $\overline{X}$ has a normal distribution with mean 1350 and standard deviation 18.97.

To determine $\beta(1350)$, we first need to figure out the threshold between P-value $\leq \alpha$ and P-value $> \alpha$ in terms of $\overline{x}$. Parallel to part (b), proceed as follows:

$.01 = P$(reject H_0 when H_0 is true) = $P(\overline{X} \geq \overline{x}$ when H_0 is true) = $1 - \Phi\left(\frac{\overline{x}-1300}{18.97}\right) \Rightarrow$

$\Phi\left(\frac{\overline{x}-1300}{18.97}\right) = .99 \Rightarrow \frac{\overline{x}-1300}{18.97} = 2.33 \Rightarrow \overline{x} = 1344.21$. That is, we'd reject H_0 at the $\alpha = .01$ level iff the observed value of $\overline{X}$ is ≥ 1344.21.

Finally, $\beta(1350) = P$(do not reject H_0 when $\mu = 1350$) = $P(\overline{X} < 1344.21$ when $\mu = 1350)$ = $P\left(Z < \frac{1344.21-1350}{18.97}\right) \approx \Phi(-.31) = .3783$.

30. The hypotheses are H_0: $\mu = 7.0$ versus H_a: $\mu < 7.0$. In each case, we want the one-tail area to the left of the observed test statistic.

a. $n = 6 \Rightarrow$ df = 6 – 1 = 5. From Table A.8, $P(T \leq -2.3$ when $T \sim t_5) = P(T \geq 2.3$ when $T \sim t_5) = .035$. Since $.035 \leq .05$, we reject H_0 at the $\alpha = .05$ level.

b. Similarly, P-value = $P(T \geq 3.1$ when $T \sim t_{14}) = .004$. Since $.004 < .01$, reject H_0.

c. Similarly, P-value = $P(T \geq 1.3$ when $T \sim t_{11}) = .110$. Since $.110 \geq .05$, do not reject H_0.

d. Here, P-value = $P(T \leq .7$ when $T \sim t_5)$ because it's a lower tailed test, and this is $1 - P(T > .7$ when $T \sim t_5) = 1 - .258 = .742$. Since $.742 > .05$, do not reject H_0. (Note: since the sign of the t-statistic contradicted H_a, we know immediately not to reject H_0.)

e. The observed value of the test statistic is $t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{6.68 - 7.0}{.0820} = -3.90$. From this, similar to parts (a)-(c), P-value = $P(T \geq 3.90$ when $T \sim t_5) = .006$ according to Table A.8. We would reject H_0 for any significance level at or above .006.

34. The data provided has $n = 6$, $\bar{x} = 31.233$, and $s = 0.689$.

a. The hypotheses are H_0: $\mu = 30$ versus H_a: $\mu > 30$. The one-sample t statistic is $t = \dfrac{31.233 - 30}{0.689/\sqrt{6}} = 4.38$.

At df = 6 – 1 = 5, the P-value is $P(T \geq 4.38) = .004$ from software. Since $.004 \leq .01$, we reject H_0 at the $\alpha = .01$ level and conclude the true average stopping distance does exceed 30 ft.

b. Using statistical software, the power of the test (with $n = 6$, $\alpha = .01$, and $\sigma = .65$) is .660 when $\mu = 31$ and .998 when $\mu = 32$. The corresponding chances of a type II error are $\beta(31) = 1 - .660 = .340$ and $\beta(32) = 1 - .998 = .002$.

c. Changing σ from .65 to .80 results in $\beta(31) = 1 - .470 = .530$ and $\beta(32) = 1 - .975 = .025$. It's still the case, as it should be, that type II error is less likely when the true value of μ is farther from μ_0. But, with the larger standard deviation, random sampling variation makes it more likely for the sample meant to deviate substantially from μ and, in particular, trick us into thinking H_0 might be plausible (even though it's false in both cases). We see β increase from .340 to .530 when $\mu = 31$ and from .002 to .025 when $\mu = 32$.

d. According to statistical software, $n = 9$ will suffice to achieve $\beta = .1$ (power = .9) when $\alpha = .01$, $\mu = 31$, and $\sigma = .65$.

38. μ = the true average percentage of organic matter in this type of soil, and the hypotheses are H_0: $\mu = 3$ versus H_a: $\mu \neq 3$. With $n = 30$, and assuming normality, we use the t test:

$t = \dfrac{\bar{x} - 3}{s/\sqrt{n}} = \dfrac{2.481 - 3}{.295} = \dfrac{-.519}{.295} = -1.759$. At df = 30 – 1 = 29, P-value = $2P(T > 1.759) = 2(.041) = .082$.

At significance level .10, since $.082 \leq .10$, we would reject H_0 and conclude that the true average percentage of organic matter in this type of soil is something other than 3. At significance level .05, we would not have rejected H_0.

40. The hypotheses are H_0: $\mu = 48$ versus H_a: $\mu > 48$. Using the one-sample t procedure, the test statistic and P-value are $t = \dfrac{51.3 - 48}{1.2/\sqrt{10}} = 8.7$ and $P(T \geq 8.7 \text{ when df} = 9) \approx 0$. Hence, we reject H_0 at any reasonable significance level and conclude that the true average strength for the WSF/cellulose composite definitely exceeds 48 MPa.